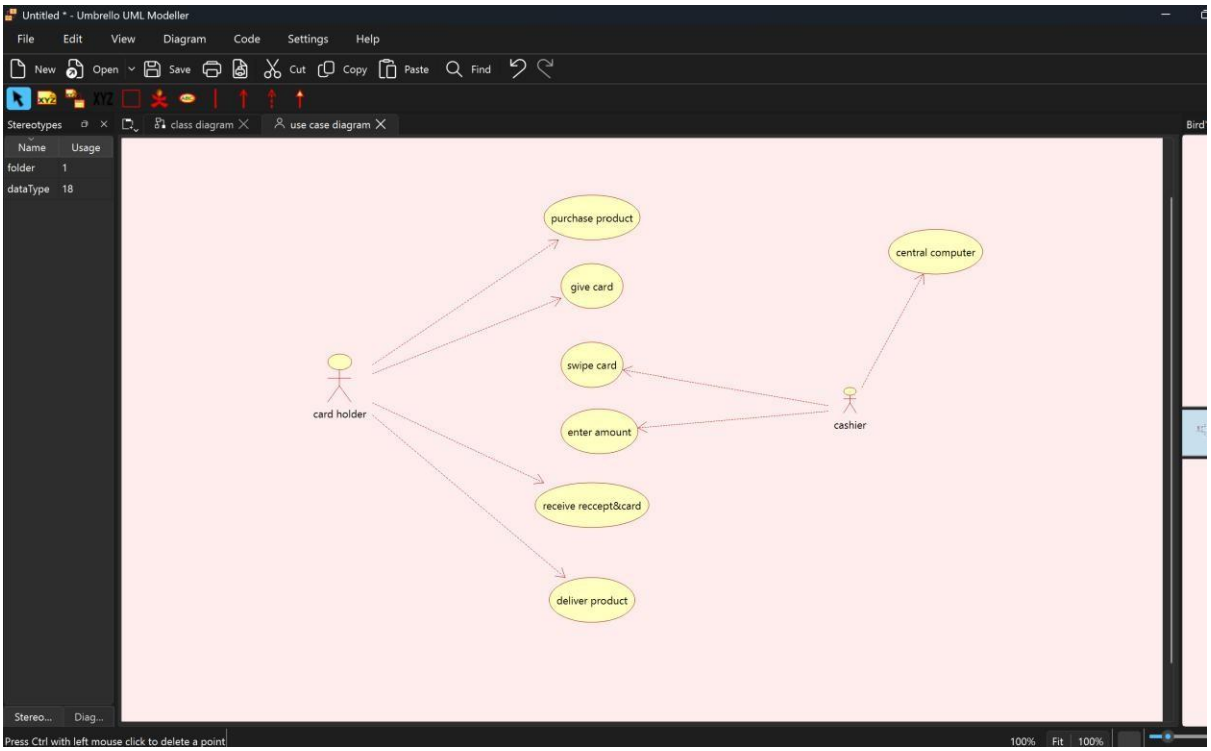
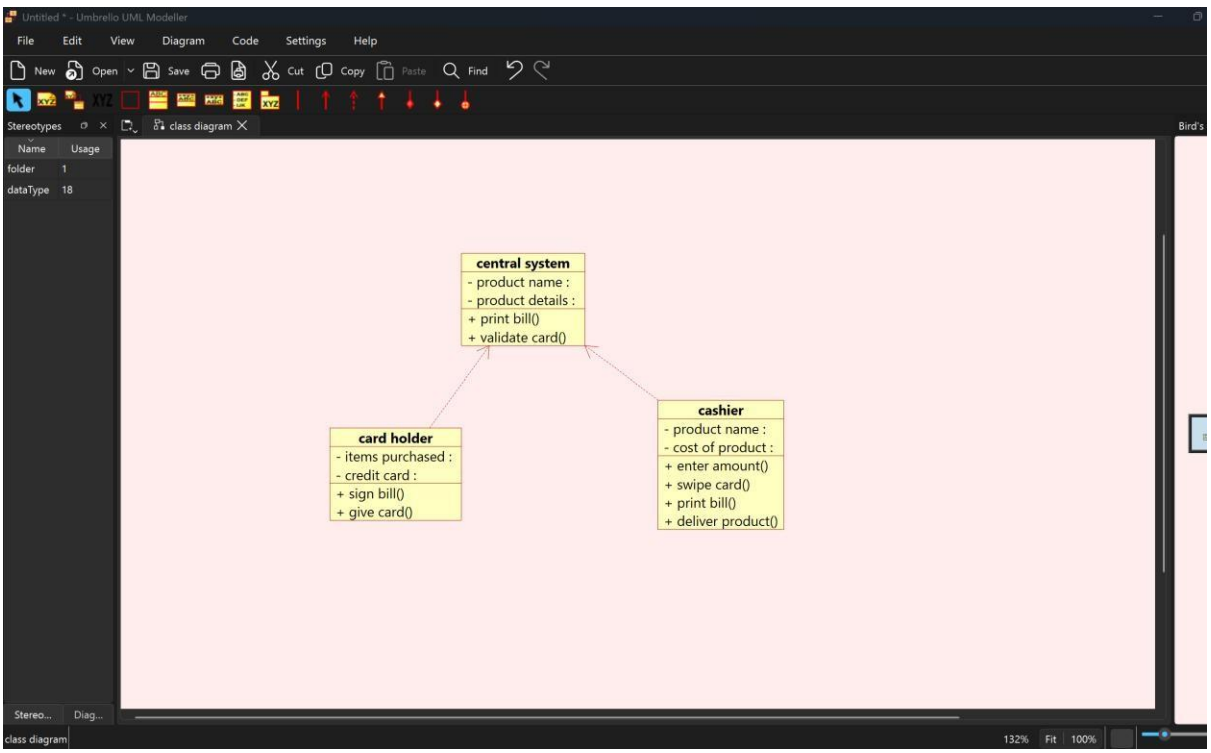
Ex No: 7 – CREDIT CARD PROCESSING SYSTEM

NAME: Arava Raja (192365033)

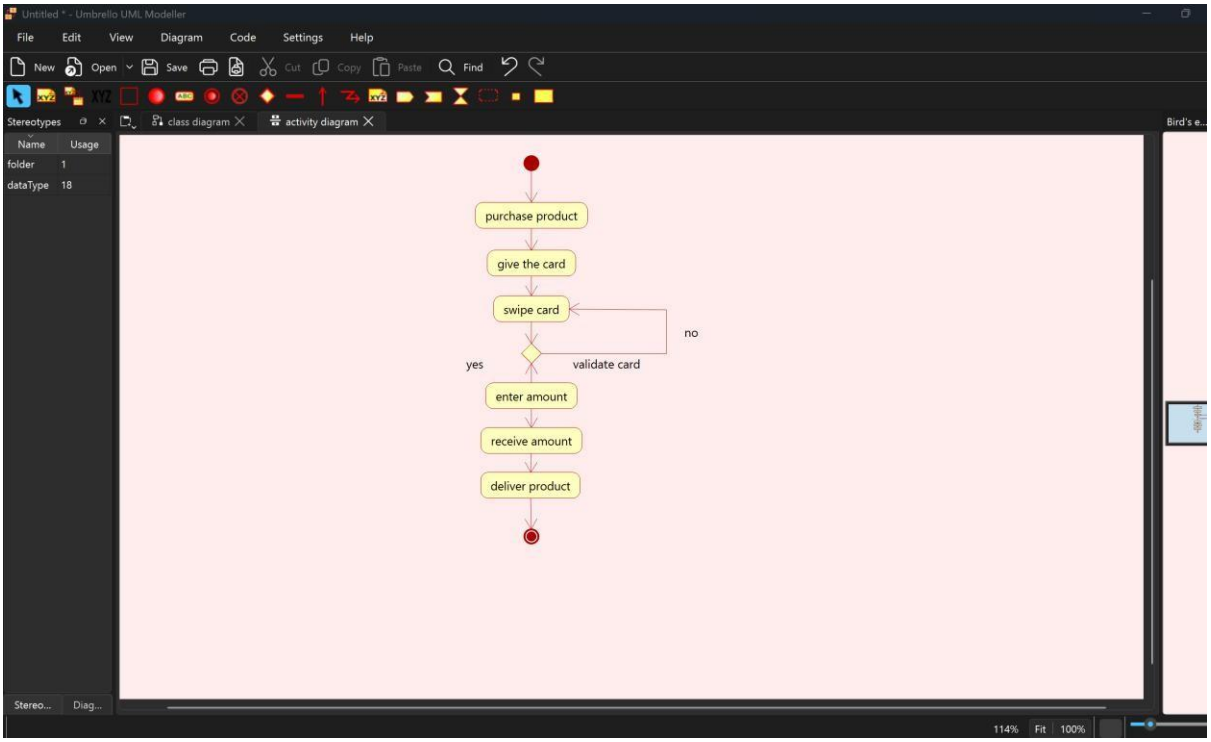
USE CASE DIAGRAM:



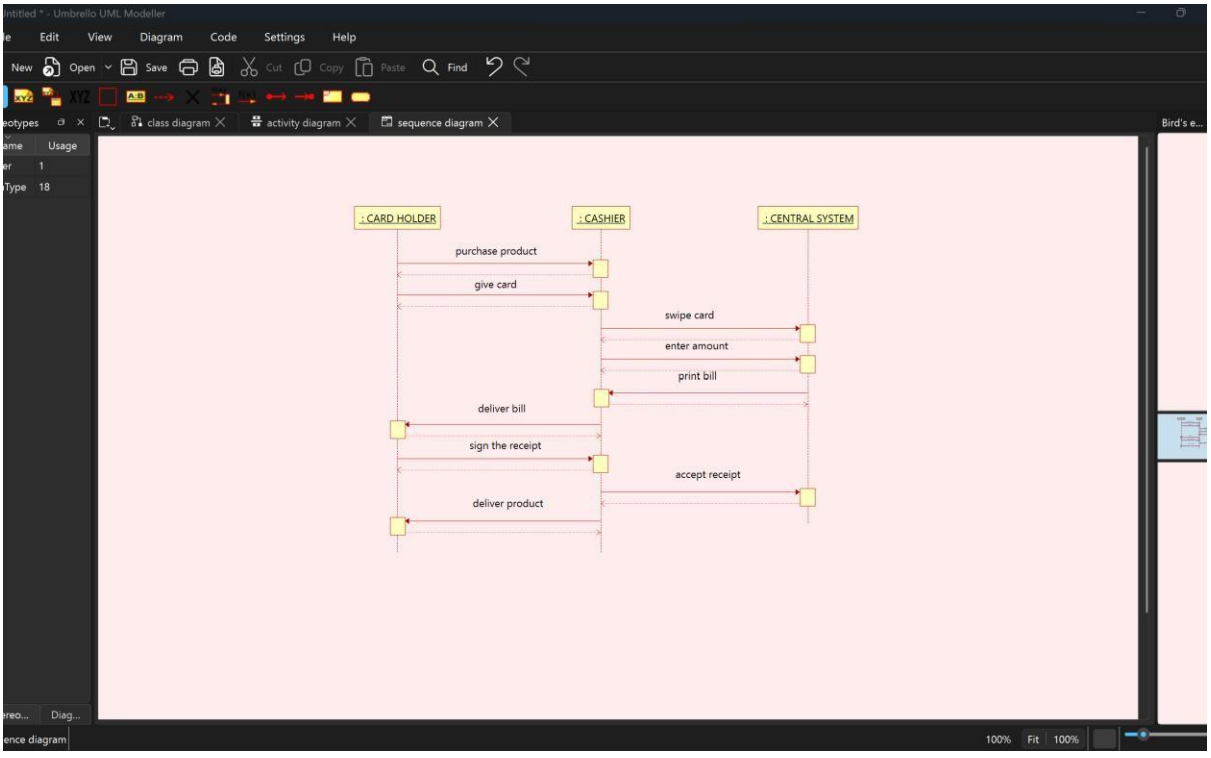
CLASS DIAGRAM:



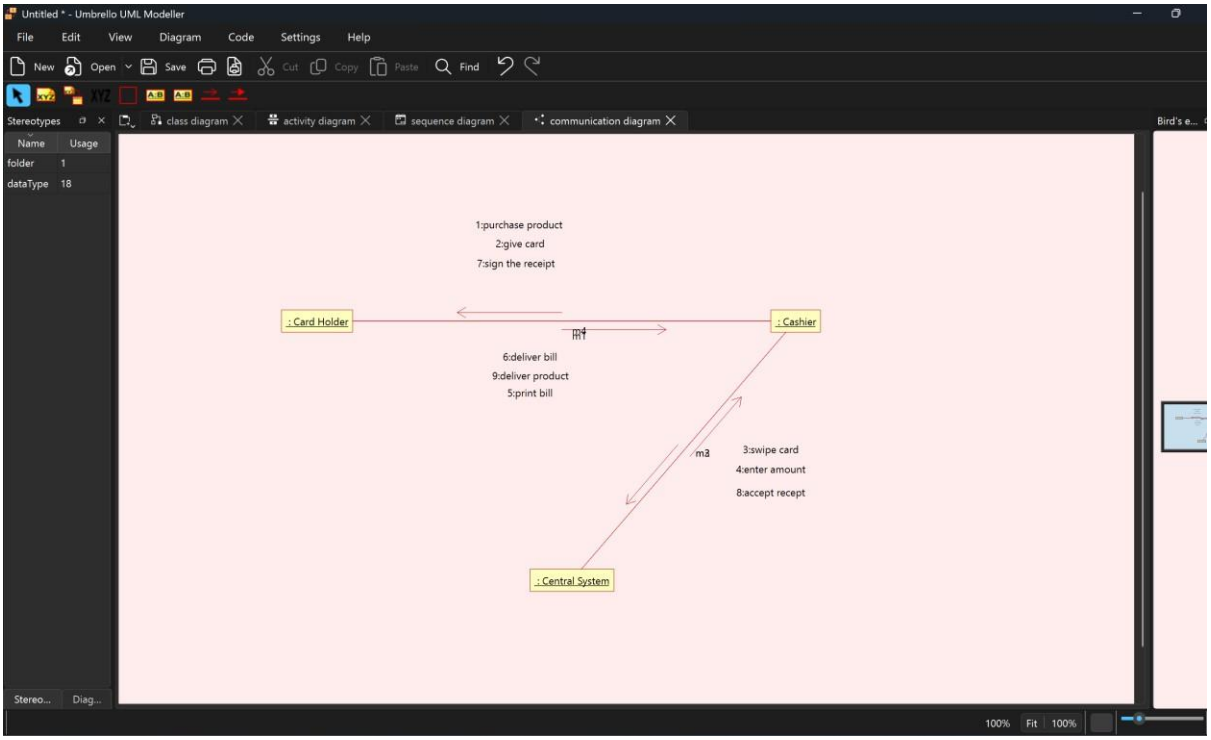
ACTIVITY DIAGRAM:



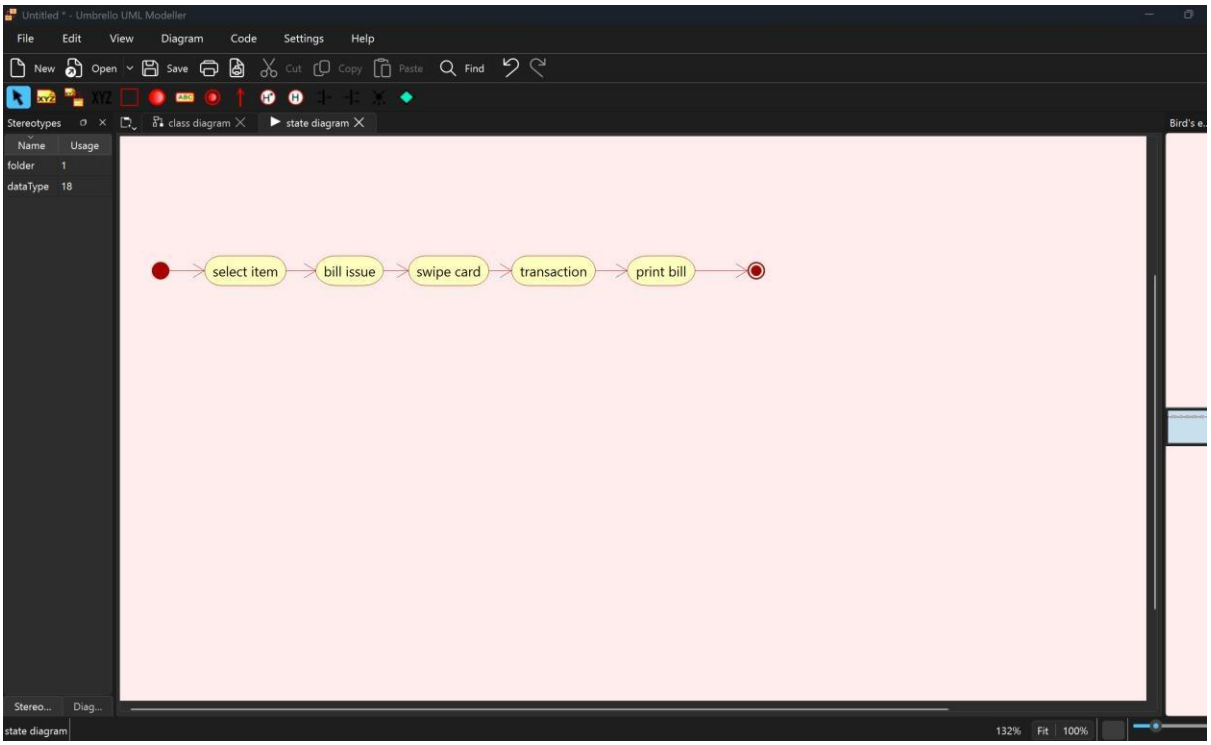
Sequence Diagram:



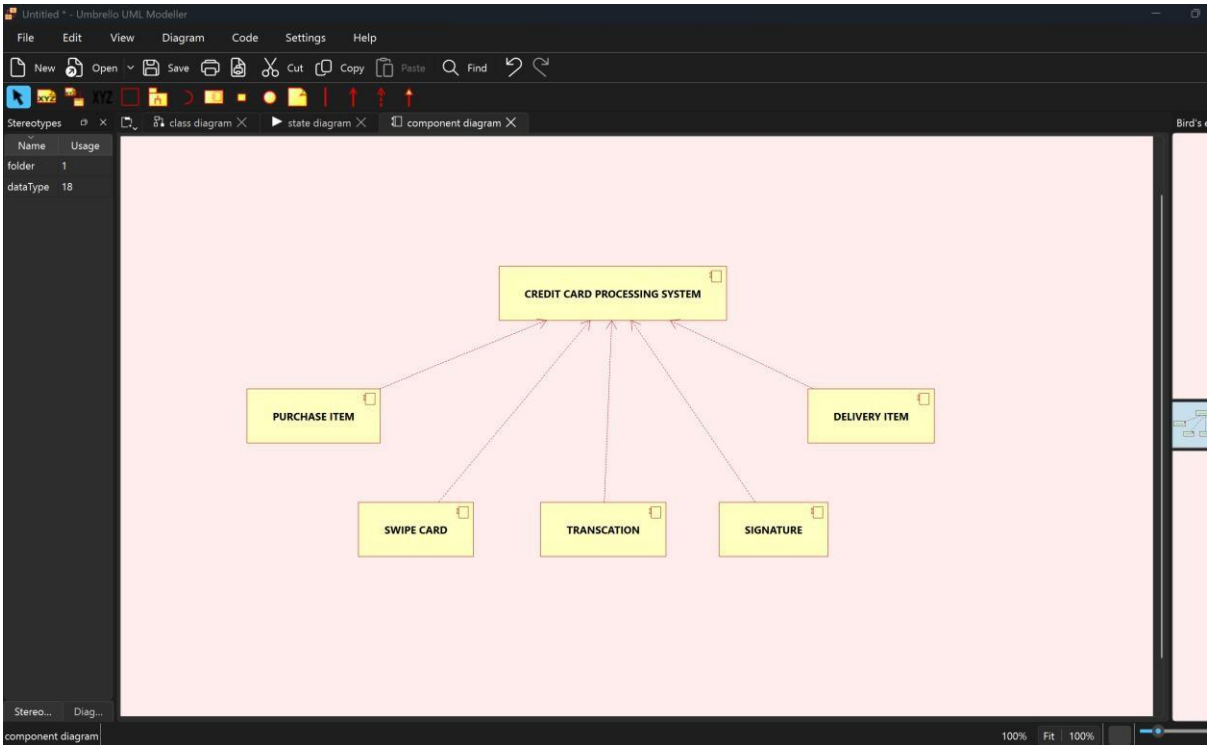
COLLABORATION DIAGRAM:



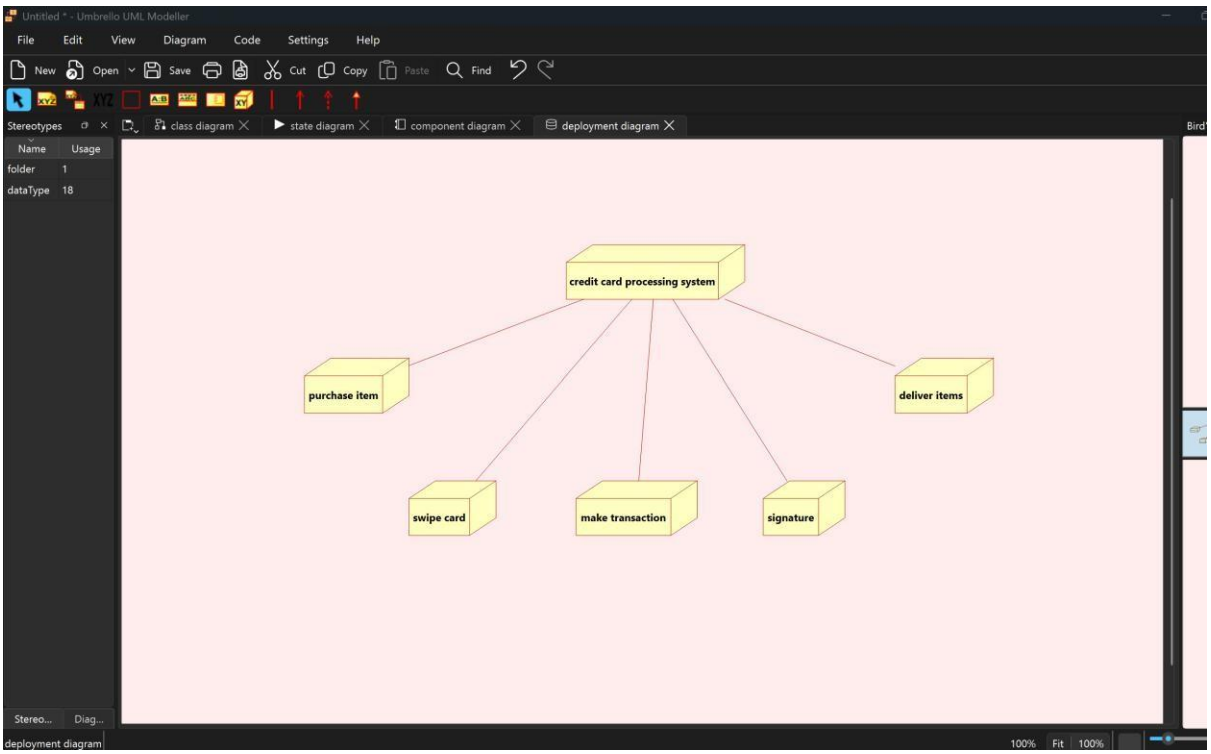
STATE CHART DIAGRAM:



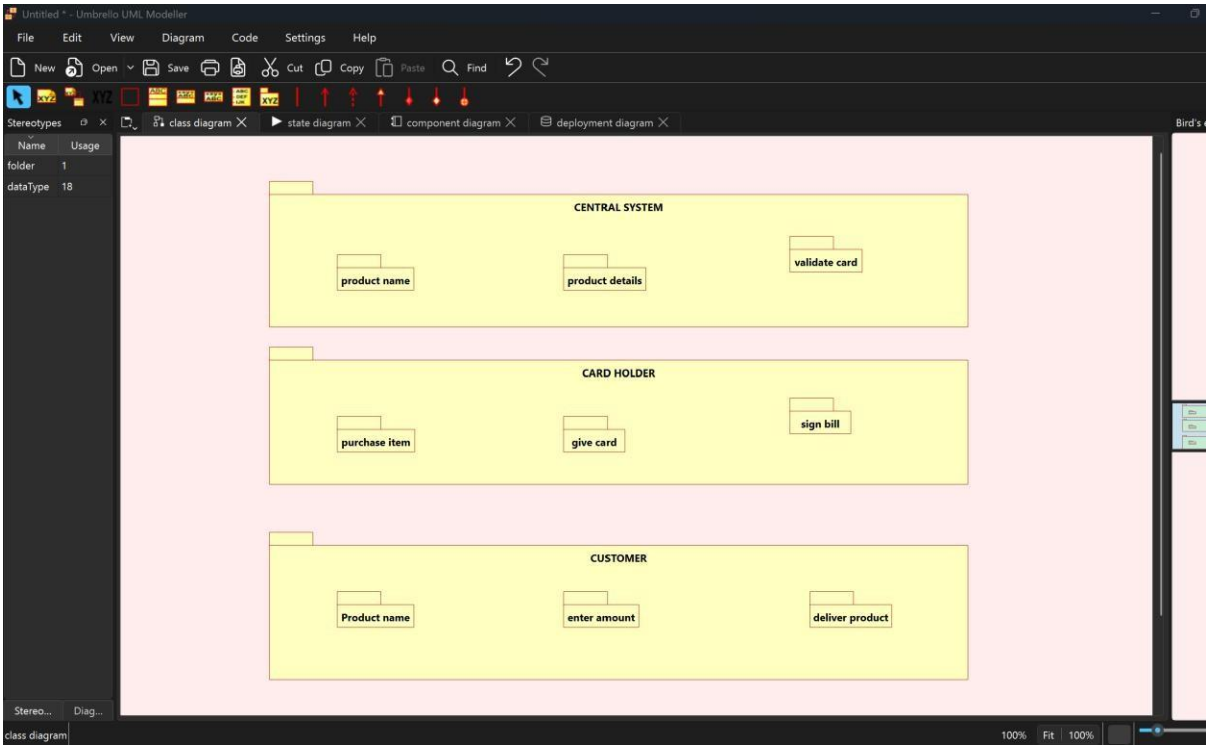
COMPONENT DIAGRAM:



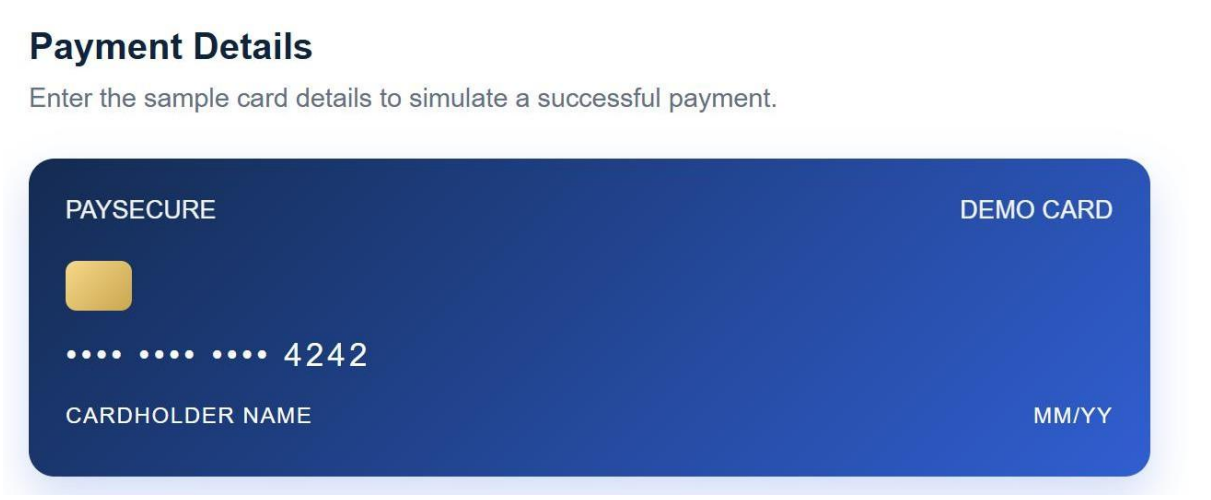
DEPLOYMENT DIAGRAM:



PACKAGE DIAGRAM:



WEBPAGE FOR CREDIT CARD PROCESSING SYSTEM



Cardholder name

e.g. Alex Kumar

Demo card number

4242 4242 4242 4242

Use the demo number: 4242 4242 4242 4242

Expiry date

MM/YY

CVV (demo only)

123

Never enter your real CVV on a demo site.

Receipt email

you@example.com

Simulate Payment

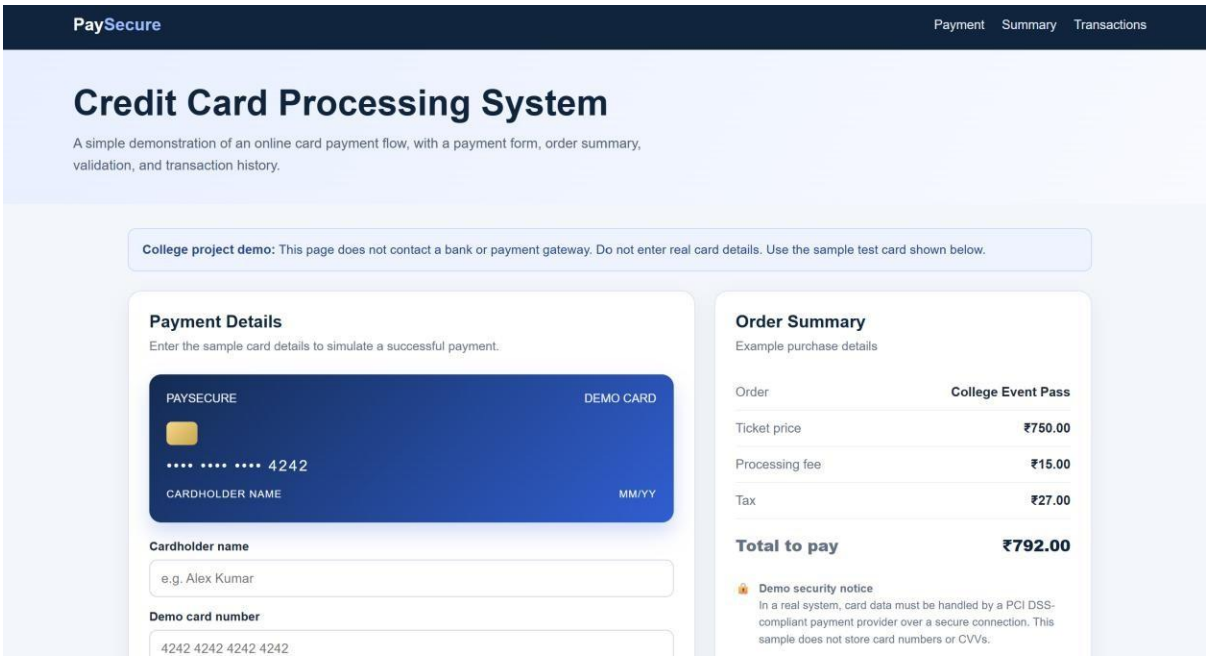
Order Summary

Example purchase details

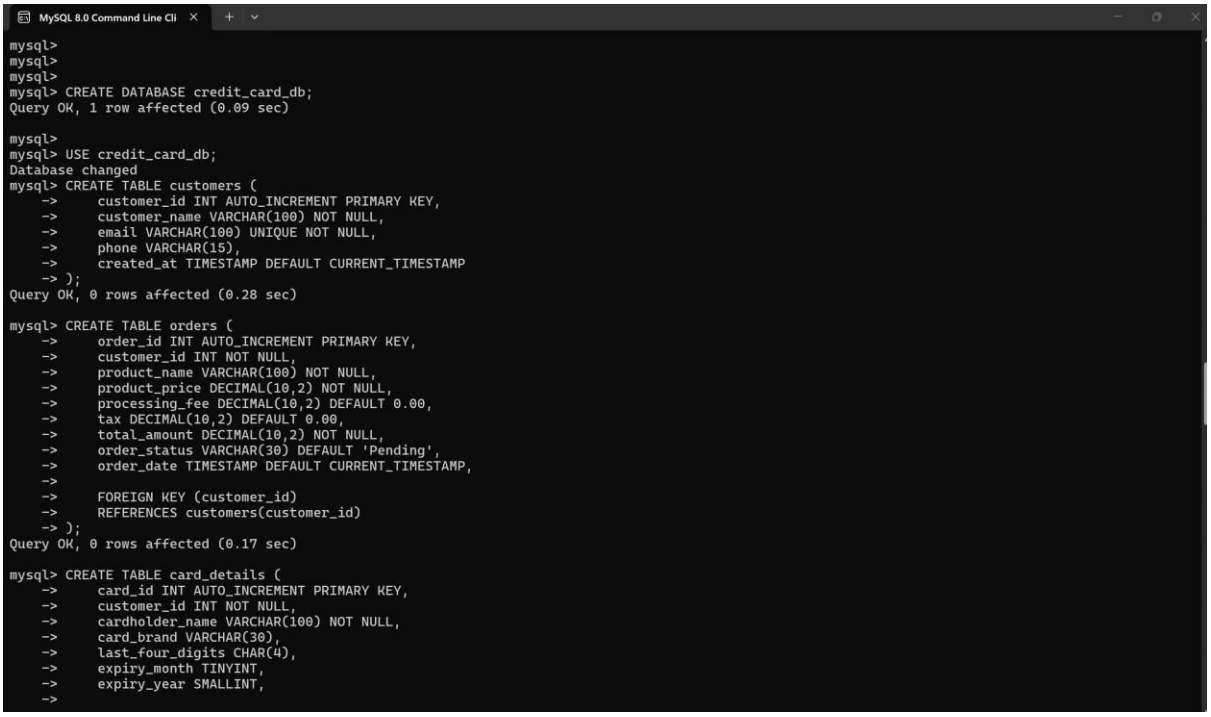
Order	**College Event Pass**
Ticket price	**₹750.00**
Processing fee	**₹15.00**
Tax	**₹27.00**
Total to pay	**₹792.00**

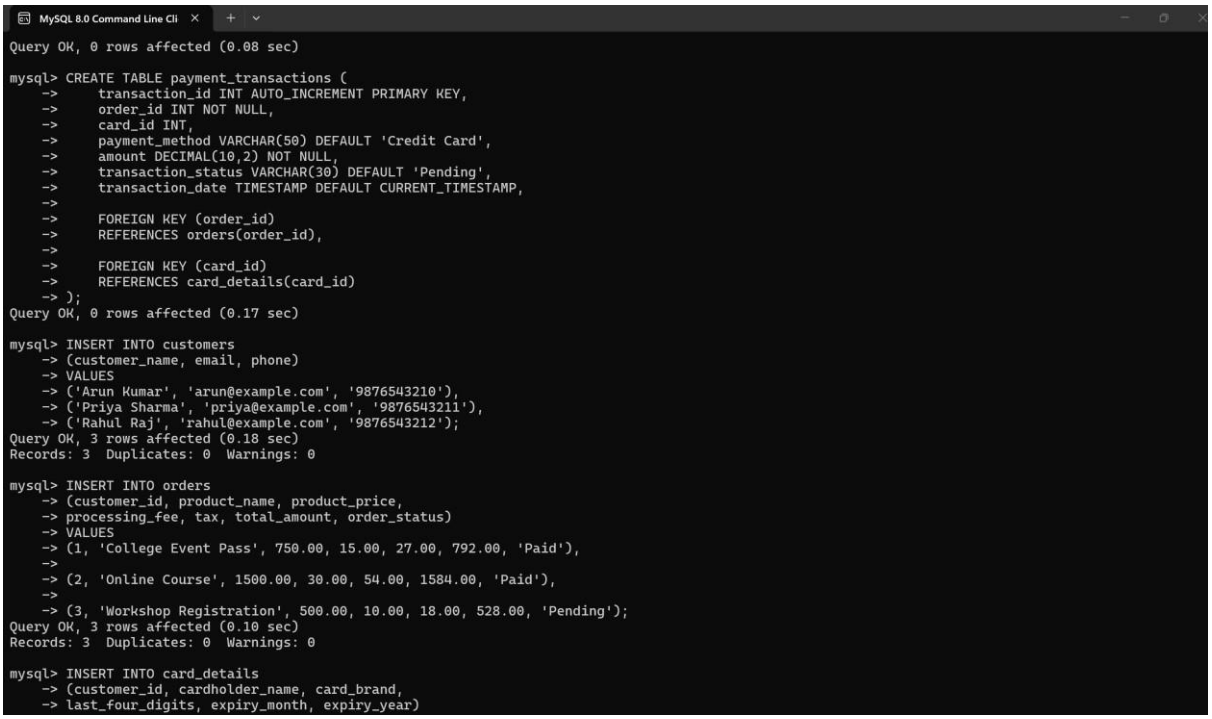
🔒 **Demo security notice**

In a real system, card data must be handled by a PCI DSS-compliant payment provider over a secure connection. This sample does not store card numbers or CVVs.



MYSQL DATABASE:





```
MySQL 8.0 Command Line Cli

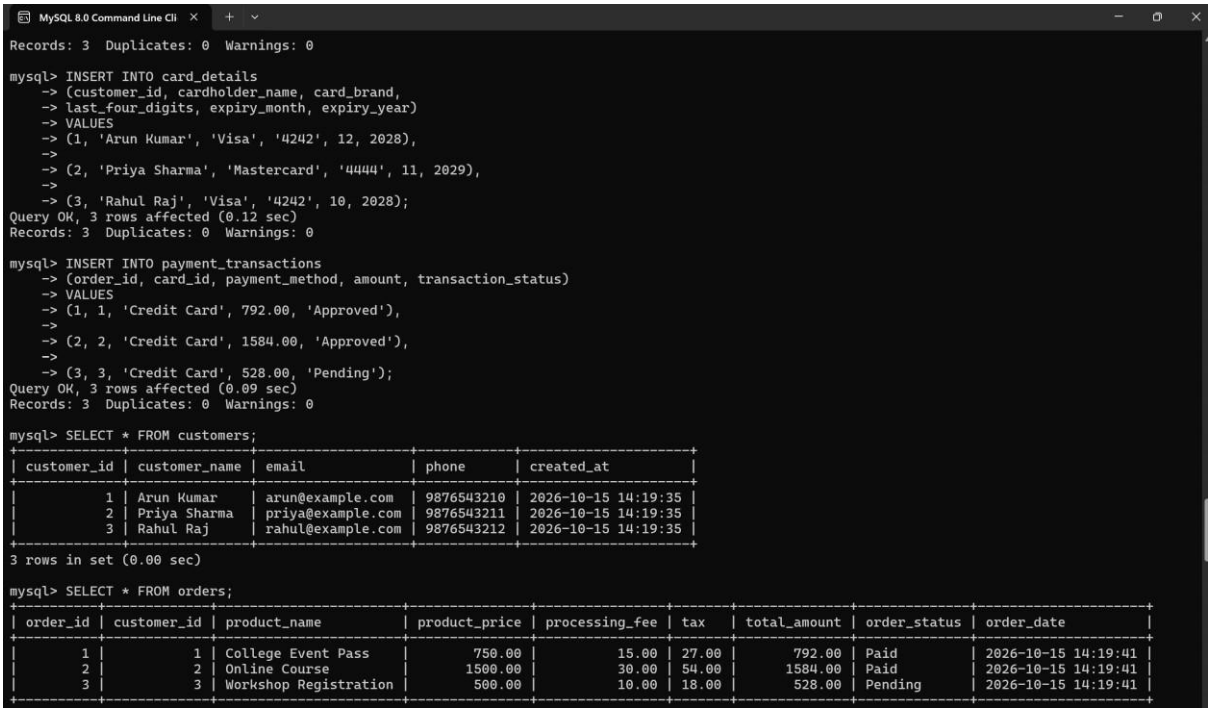
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE payment_transactions (
    ->     transaction_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     order_id INT NOT NULL,
    ->     card_id INT,
    ->     payment_method VARCHAR(50) DEFAULT 'Credit Card',
    ->     amount DECIMAL(10,2) NOT NULL,
    ->     transaction_status VARCHAR(30) DEFAULT 'Pending',
    ->     transaction_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     FOREIGN KEY (order_id)
    ->     REFERENCES orders(order_id),
    ->
    ->     FOREIGN KEY (card_id)
    ->     REFERENCES card_details(card_id)
    -> );
Query OK, 0 rows affected (0.17 sec)

mysql> INSERT INTO customers
    -> (customer_name, email, phone)
    -> VALUES
    -> ('Arun Kumar', 'arun@example.com', '9876543210'),
    -> ('Priya Sharma', 'priya@example.com', '9876543211'),
    -> ('Rahul Raj', 'rahul@example.com', '9876543212');
Query OK, 3 rows affected (0.18 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO orders
    -> (customer_id, product_name, product_price,
    -> processing_fee, tax, total_amount, order_status)
    -> VALUES
    -> (1, 'College Event Pass', 750.00, 15.00, 27.00, 792.00, 'Paid'),
    ->
    -> (2, 'Online Course', 1500.00, 30.00, 54.00, 1584.00, 'Paid'),
    ->
    -> (3, 'Workshop Registration', 500.00, 10.00, 18.00, 528.00, 'Pending');
Query OK, 3 rows affected (0.10 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO card_details
    -> (customer_id, cardholder_name, card_brand,
    -> last_four_digits, expiry_month, expiry_year)
```



```
MySQL 8.0 Command Line Cli

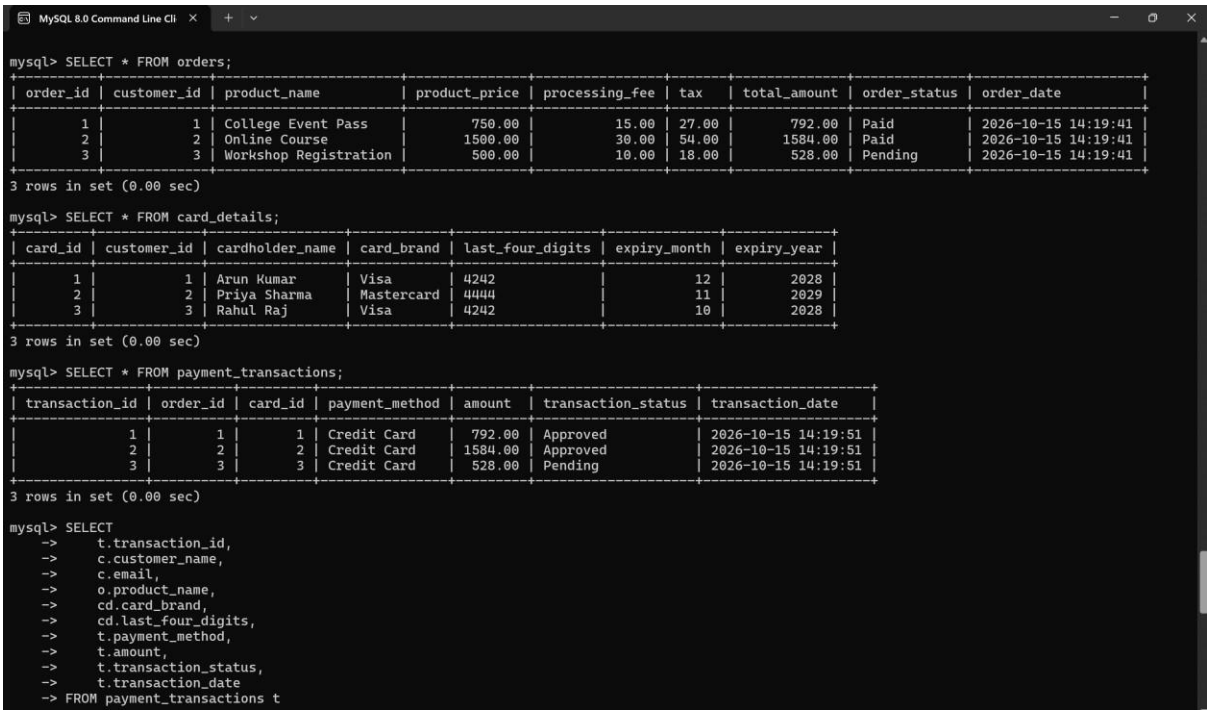
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO card_details
    -> (customer_id, cardholder_name, card_brand,
    -> last_four_digits, expiry_month, expiry_year)
    -> VALUES
    -> (1, 'Arun Kumar', 'Visa', '4242', 12, 2028),
    ->
    -> (2, 'Priya Sharma', 'Mastercard', '4444', 11, 2029),
    ->
    -> (3, 'Rahul Raj', 'Visa', '4242', 10, 2028);
Query OK, 3 rows affected (0.12 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO payment_transactions
    -> (order_id, card_id, payment_method, amount, transaction_status)
    -> VALUES
    -> (1, 1, 'Credit Card', 792.00, 'Approved'),
    ->
    -> (2, 2, 'Credit Card', 1584.00, 'Approved'),
    ->
    -> (3, 3, 'Credit Card', 528.00, 'Pending');
Query OK, 3 rows affected (0.09 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM customers;
+-------------+---------------+-------------------+------------+---------------------+
| customer_id | customer_name | email             | phone      | created_at          |
+-------------+---------------+-------------------+------------+---------------------+
|           1 | Arun Kumar    | arun@example.com  | 9876543210 | 2026-10-15 14:19:35 |
|           2 | Priya Sharma  | priya@example.com | 9876543211 | 2026-10-15 14:19:35 |
|           3 | Rahul Raj     | rahul@example.com | 9876543212 | 2026-10-15 14:19:35 |
+-------------+---------------+-------------------+------------+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM orders;
+----------+-------------+----------------------+---------------+----------------+-------+--------------+--------------+---------------------+
| order_id | customer_id | product_name         | product_price | processing_fee | tax   | total_amount | order_status | order_date          |
+----------+-------------+----------------------+---------------+----------------+-------+--------------+--------------+---------------------+
|        1 |           1 | College Event Pass   |        750.00 |          15.00 | 27.00 |       792.00 | Paid         | 2026-10-15 14:19:41 |
|        2 |           2 | Online Course        |       1500.00 |          30.00 | 54.00 |      1584.00 | Paid         | 2026-10-15 14:19:41 |
|        3 |           3 | Workshop Registration |        500.00 |          10.00 | 18.00 |       528.00 | Pending      | 2026-10-15 14:19:41 |
+----------+-------------+----------------------+---------------+----------------+-------+--------------+--------------+---------------------+
```



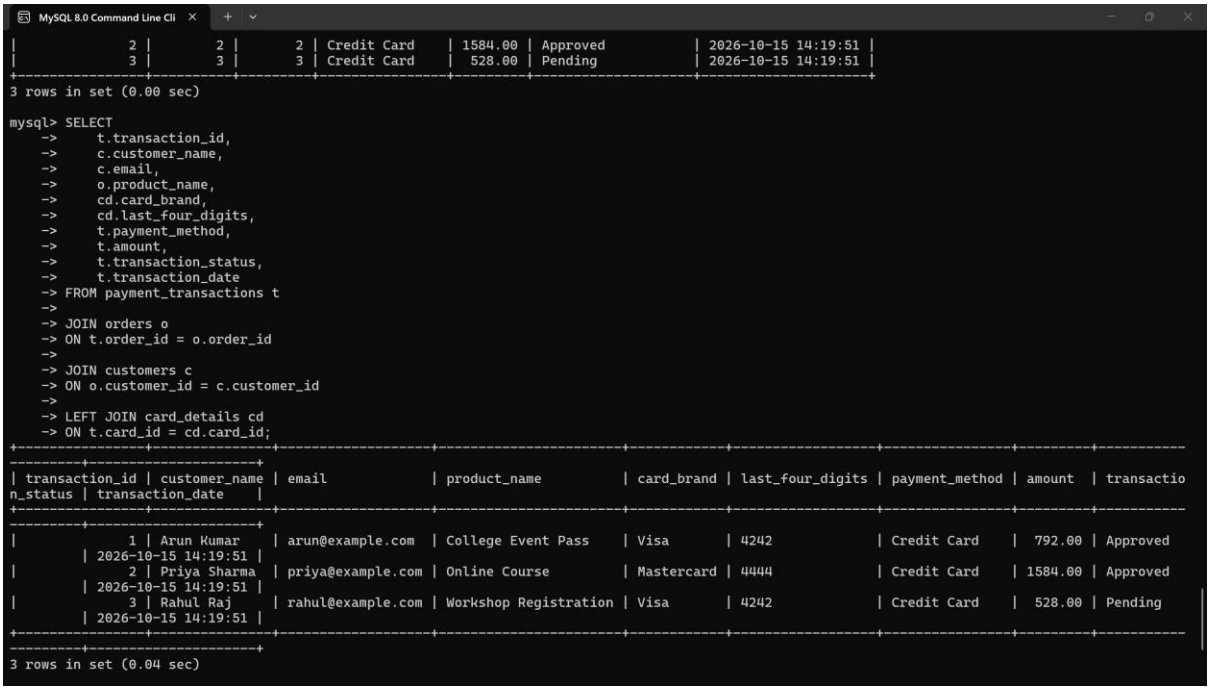
```
MySQL 8.0 Command Line Cli

mysql> SELECT * FROM orders;
+----------+-------------+----------------------+---------------+----------------+-------+--------------+--------------+---------------------+
| order_id | customer_id | product_name         | product_price | processing_fee | tax   | total_amount | order_status | order_date          |
+----------+-------------+----------------------+---------------+----------------+-------+--------------+--------------+---------------------+
|        1 |           1 | College Event Pass   |        750.00 |          15.00 | 27.00 |       792.00 | Paid         | 2026-10-15 14:19:41 |
|        2 |           2 | Online Course        |       1500.00 |          30.00 | 54.00 |      1584.00 | Paid         | 2026-10-15 14:19:41 |
|        3 |           3 | Workshop Registration |        500.00 |          10.00 | 18.00 |       528.00 | Pending      | 2026-10-15 14:19:41 |
+----------+-------------+----------------------+---------------+----------------+-------+--------------+--------------+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM card_details;
+---------+-------------+-----------------+------------+------------------+--------------+-------------+
| card_id | customer_id | cardholder_name | card_brand | last_four_digits | expiry_month | expiry_year |
+---------+-------------+-----------------+------------+------------------+--------------+-------------+
|       1 |           1 | Arun Kumar      | Visa       | 4242             |           12 |        2028 |
|       2 |           2 | Priya Sharma    | Mastercard | 4444             |           11 |        2029 |
|       3 |           3 | Rahul Raj       | Visa       | 4242             |           10 |        2028 |
+---------+-------------+-----------------+------------+------------------+--------------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM payment_transactions;
+----------------+----------+---------+----------------+---------+--------------------+---------------------+
| transaction_id | order_id | card_id | payment_method | amount  | transaction_status | transaction_date    |
+----------------+----------+---------+----------------+---------+--------------------+---------------------+
|              1 |        1 |       1 | Credit Card    |  792.00 | Approved           | 2026-10-15 14:19:51 |
|              2 |        2 |       2 | Credit Card    | 1584.00 | Approved           | 2026-10-15 14:19:51 |
|              3 |        3 |       3 | Credit Card    |  528.00 | Pending            | 2026-10-15 14:19:51 |
+----------------+----------+---------+----------------+---------+--------------------+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT
    ->     t.transaction_id,
    ->     c.customer_name,
    ->     c.email,
    ->     o.product_name,
    ->     cd.card_brand,
    ->     cd.last_four_digits,
    ->     t.payment_method,
    ->     t.amount,
    ->     t.transaction_status,
    ->     t.transaction_date
    -> FROM payment_transactions t
```

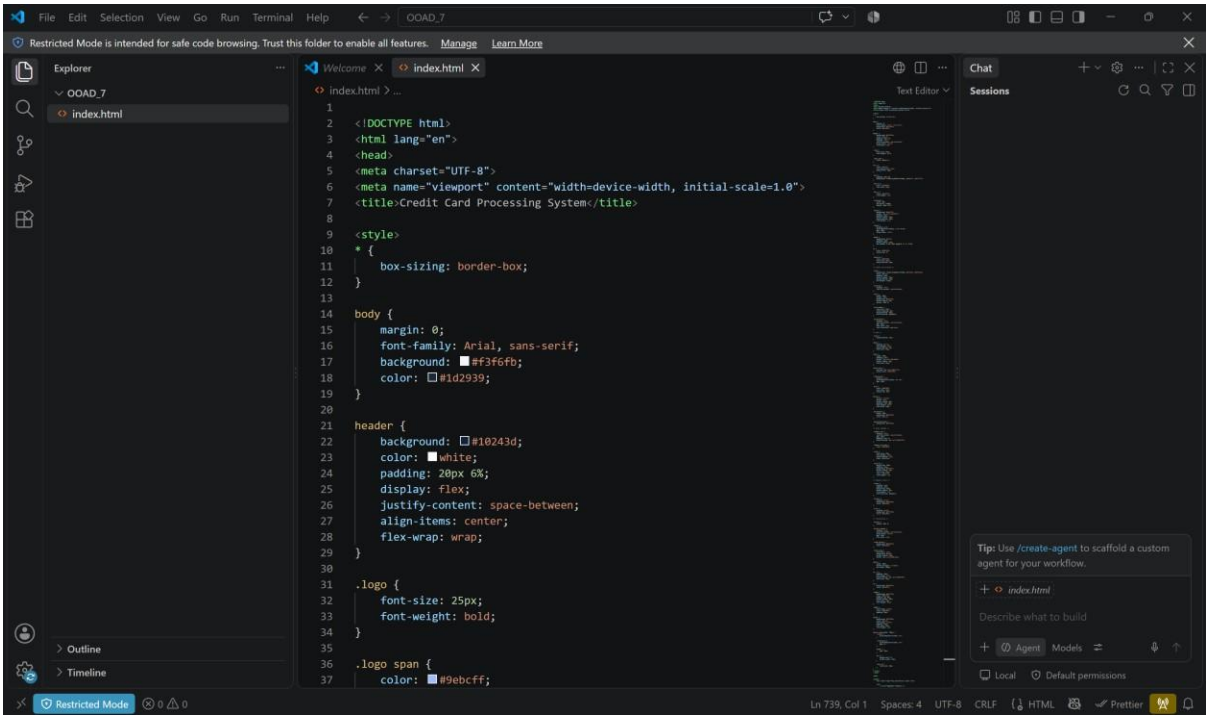


```
MySQL 8.0 Command Line Cli

|              2 |        2 |       2 | Credit Card    | 1584.00 | Approved           | 2026-10-15 14:19:51 |
|              3 |        3 |       3 | Credit Card    |  528.00 | Pending            | 2026-10-15 14:19:51 |
+----------------+----------+---------+----------------+---------+--------------------+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT
    ->     t.transaction_id,
    ->     c.customer_name,
    ->     c.email,
    ->     o.product_name,
    ->     cd.card_brand,
    ->     cd.last_four_digits,
    ->     t.payment_method,
    ->     t.amount,
    ->     t.transaction_status,
    ->     t.transaction_date
    -> FROM payment_transactions t
    ->
    -> JOIN orders o
    -> ON t.order_id = o.order_id
    ->
    -> JOIN customers c
    -> ON o.customer_id = c.customer_id
    ->
    -> LEFT JOIN card_details cd
    -> ON t.card_id = cd.card_id;
+----------------+---------------+-------------------+----------------------+------------+------------------+----------------+---------+-----------
---------+---------------------+
| transaction_id | customer_name | email             | product_name         | card_brand | last_four_digits | payment_method | amount  | transactio
n_status | transaction_date    |
+----------------+---------------+-------------------+----------------------+------------+------------------+----------------+---------+-----------
---------+---------------------+
|              1 | Arun Kumar    | arun@example.com  | College Event Pass   | Visa       | 4242             | Credit Card    |  792.00 | Approved
       | 2026-10-15 14:19:51 |
|              2 | Priya Sharma  | priya@example.com | Online Course        | Mastercard | 4444             | Credit Card    | 1584.00 | Approved
       | 2026-10-15 14:19:51 |
|              3 | Rahul Raj     | rahul@example.com | Workshop Registration | Visa       | 4242             | Credit Card    |  528.00 | Pending
       | 2026-10-15 14:19:51 |
+----------------+---------------+-------------------+----------------------+------------+------------------+----------------+---------+-----------
---------+---------------------+
3 rows in set (0.04 sec)
```

HTML CODE:



```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Credit Card Processing System</title>

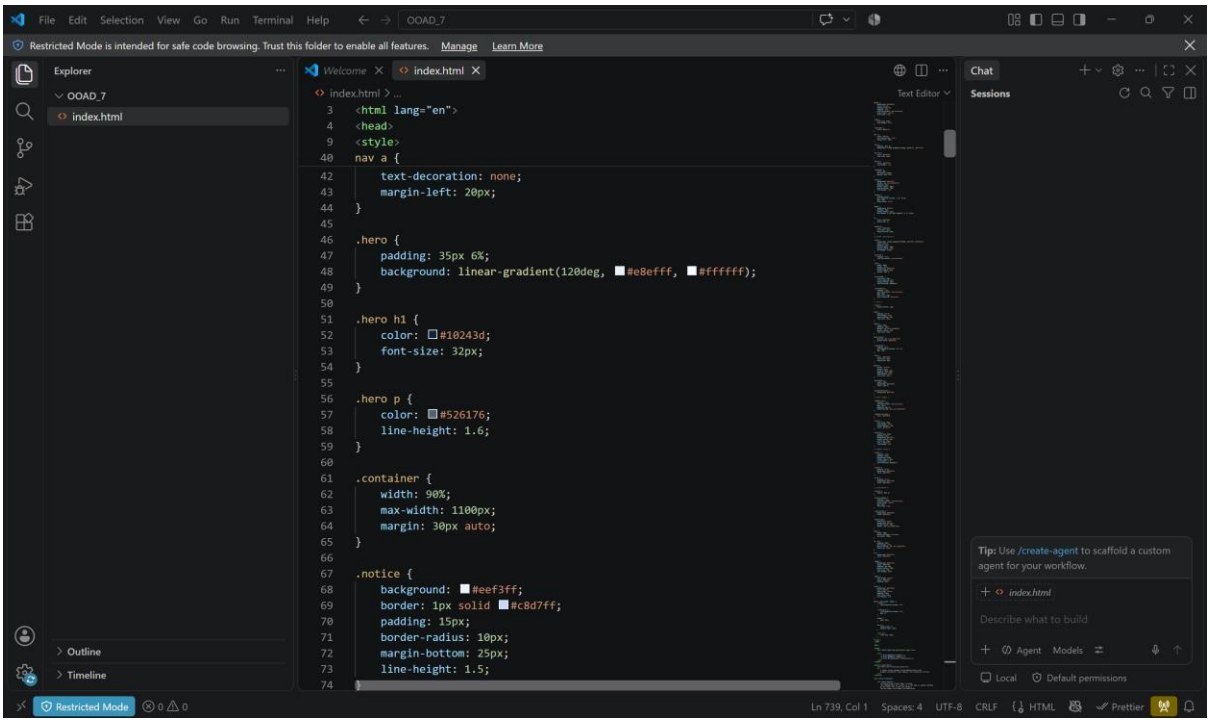
<style>
* {
    box-sizing: border-box;
}

body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: #f3f6fb;
    color: #1d2939;
}

header {
    background: #10243d;
    color: white;
    padding: 20px 6%;
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
}

.logo {
    font-size: 25px;
    font-weight: bold;
}

.logo span {
    color: #9ebcff;
```



```
<html lang="en">
<head>
<style>
nav a {
    text-decoration: none;
    margin-left: 20px;
}

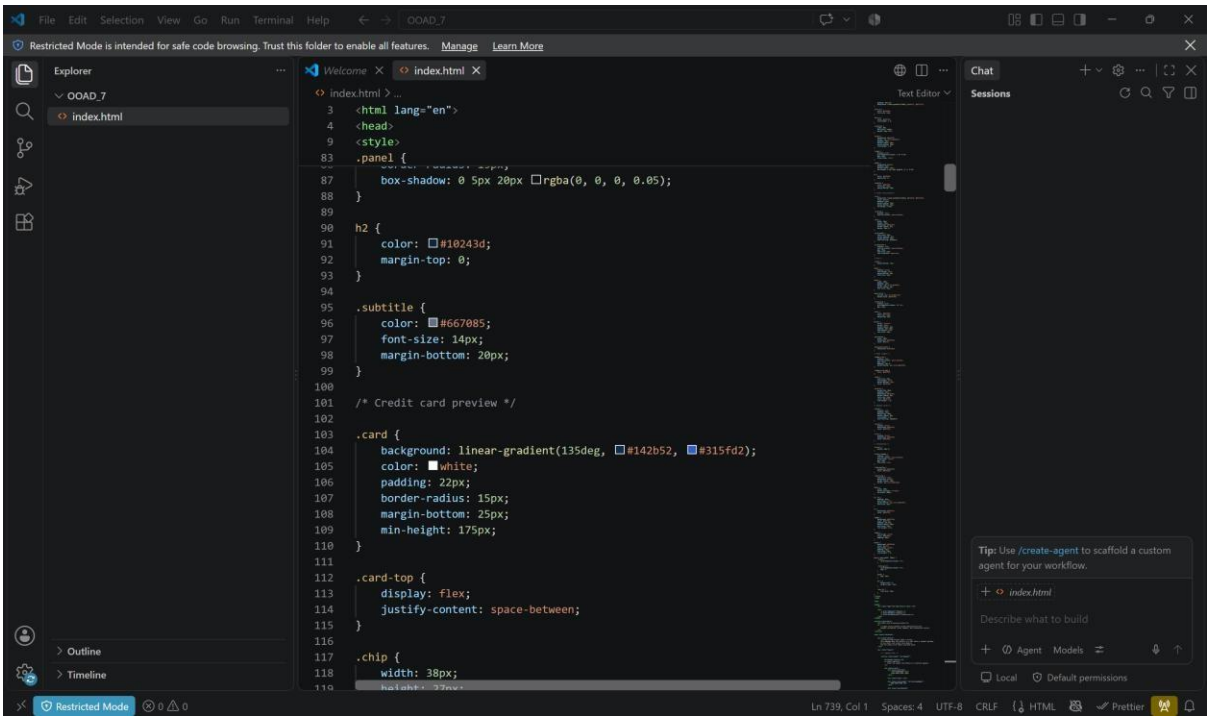
.hero {
    padding: 35px 6%;
    background: linear-gradient(120deg, #e8efff, #ffffff);
}

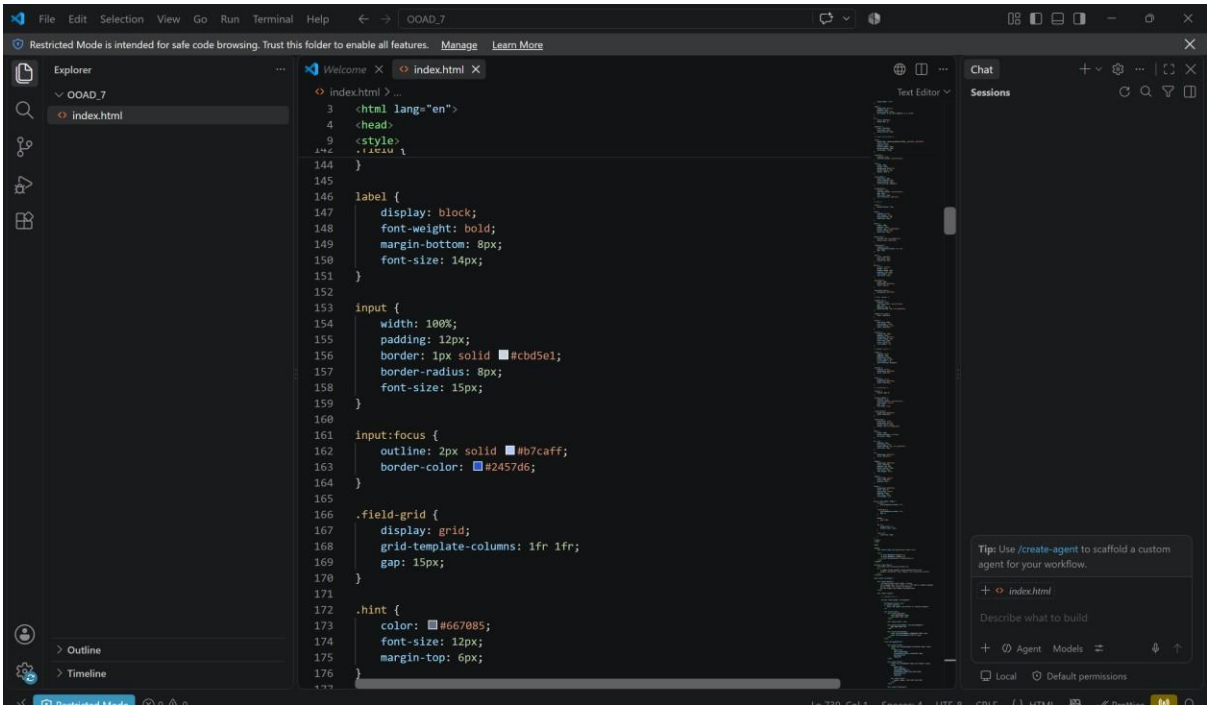
.hero h1 {
    color: #10243d;
    font-size: 32px;
}

.hero p {
    color: #526176;
    line-height: 1.6;
}

.container {
    width: 90%;
    max-width: 1100px;
    margin: 30px auto;
}

.notice {
    background: #eef3ff;
    border: 1px solid #c8d7ff;
    padding: 15px;
    border-radius: 10px;
    margin-bottom: 25px;
    line-height: 1.5;
}
```


File Edit Selection View Go Run Terminal Help
OOAD_7
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More
Explorer
OOAD_7
index.html
Welcome
index.html
index.html > ...
Text Editor
Chat
Sessions
3 <html lang="en">
4 <head>
9 <style>
83 .panel {
87 box-shadow: 0 5px 20px rgba(0, 0, 0, 0.05);
88 }
90 h2 {
91 color: #10243d;
92 margin-top: 0;
93 }
95 .subtitle {
96 color: #667085;
97 font-size: 14px;
98 margin-bottom: 20px;
99 }
101 /* Credit card preview */
103 .card {
104 background: linear-gradient(135deg, #142b52, #315fd2);
105 color: white;
106 padding: 22px;
107 border-radius: 15px;
108 margin-bottom: 25px;
109 min-height: 175px;
110 }
112 .card-top {
113 display: flex;
114 justify-content: space-between;
115 }
117 .chip {
118 width: 38px;
Tip: Use /create-agent to scaffold a custom agent for your workflow.
index.html
Describe what to build
Agent Models
Local Default permissions
Outline
Timeline
Restricted Mode
Ln 739, Col 1 Spaces: 4 UTF-8 CRLF HTML Prettier


File Edit Selection View Go Run Terminal Help
OOAD_7
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More
Explorer
OOAD_7
index.html
Welcome
index.html
index.html > ...
Text Editor
Chat
Sessions
3 <html lang="en">
4 <head>
9 <style>
144 }
146 label {
147 display: block;
148 font-weight: bold;
149 margin-bottom: 8px;
150 font-size: 14px;
151 }
153 input {
154 width: 100%;
155 padding: 12px;
156 border: 1px solid #cbd5e1;
157 border-radius: 8px;
158 font-size: 15px;
159 }
161 input:focus {
162 outline: 2px solid #b7caff;
163 border-color: #2457d6;
164 }
166 .field-grid {
167 display: grid;
168 grid-template-columns: 1fr 1fr;
169 gap: 15px;
170 }
172 .hint {
173 color: #667085;
174 font-size: 12px;
175 margin-top: 6px;
176 }
Tip: Use /create-agent to scaffold a custom agent for your workflow.
index.html
Describe what to build
Agent Models
Local Default permissions
Outline
Timeline